

Quick Start Guide: QArm Mini

STEP 1 Check Components and Details

Make sure you have the following items ready before you begin:



Ensure your QArm Mini Manipulator includes the following components

1. QArm Mini
2. Base Plate
3. 24V 2.71A Power Supply
4. USB-A to USB-C 3.0 QArm Mini interface cable

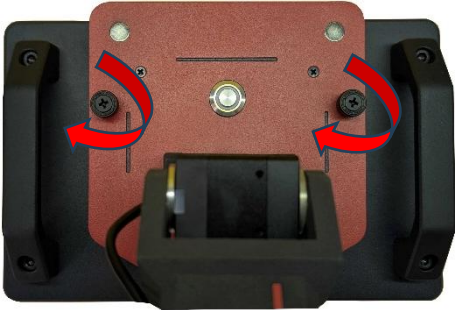
Ensure your local computer has the following components

1. Windows 10/11 operating system
2. MATLAB® 2024a or later installed w/ MATLAB® Coder & Simulink® Coder
3. QUARC 2024 SP1 or later installed

STEP 2 Setup the Hardware

The steps below outline the instructions to setup the QBot Platform for testing:

A Firmly secure the QArm Mini to the base plate by tightening the thumb screws.



B Place the QArm Mini on a flat surface and ensure that a cylindrical space of 0.5m radius and height is around it so that all the joints can rotate freely.

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1. Connect the provided power supply to the power port of the QArm Mini.
2. Connect the QArm Mini to your workstation using the provided USB-A to USB-C cable.
3. Click the power switch to turn ON the QArm Mini (only after #1 and #2 are connected).

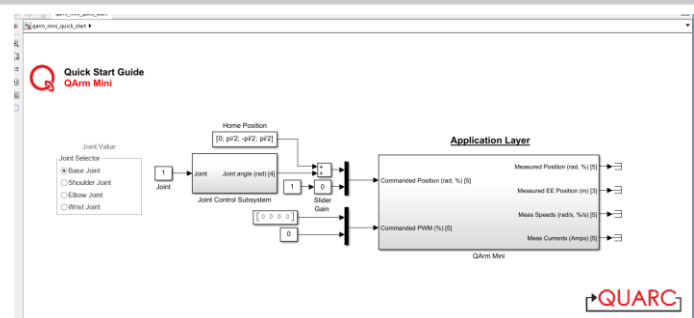
1. Ensure that the power switch is OFF.
2. Move the QArm Mini to Rest Configuration.
3. Turn ON the power switch.
4. The LED around the power switch should glow RED.

The steps below outline the instructions to run the Quick Start Example for MATLAB®/Simulink®:

Check and update the latency setting:

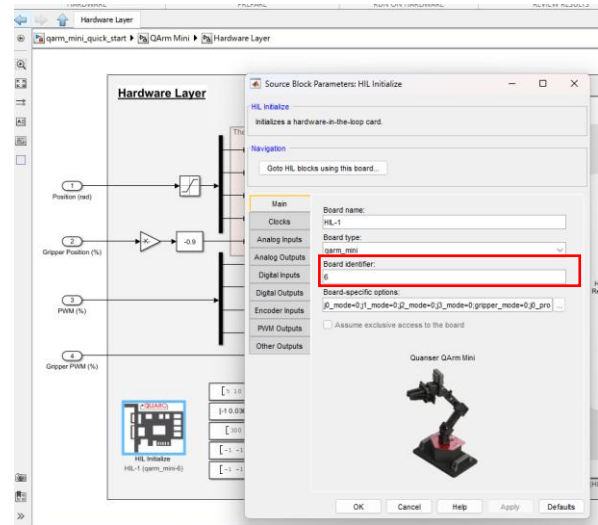
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- The screenshot displays the Windows Device Manager interface. In the left-hand pane, the 'Ports (COM & LPT)' category is expanded, showing a list of communication ports. The 'USB Serial Port (COM4)' is highlighted. The right-hand pane shows the 'USB Serial Port (COM4) Properties' dialog box with the 'General' tab selected. This tab displays the device name 'USB Serial Port (COM4)' and its location 'Universal Serial Bus controllers'. The 'Port Settings' tab is also visible, showing the following configuration: Baud rate: 9600, Data bits: 8, Parity: None, Stop bits: 1, and Flow control: None. Below the 'Port Settings' tab, the 'Advanced Settings for COM4' dialog box is open. This dialog box has several sections: 'COM Port Number' is set to 'COM4'; 'USB Transfer Sizes' includes a note to 'Select lower settings to correct performance problems at low baud rates' and 'Select higher settings for faster performance'; 'Receive (Bytes)' and 'Transmit (Bytes)' are both set to '4096'; 'BM Options' includes a note to 'Select lower settings to correct response problems' and a 'Latency Timer (msec)' set to '2'; and 'Miscellaneous Options' includes checkboxes for 'Serial Enumerator' (checked), 'Serial Printer', 'Cancel If Power Off', 'Event On Surprise Removal', and 'Set RTS On Close'.

1. Launch MATLAB.
2. Navigate to the 2_quick_start_guides directory inside the Quanser folder.
3. Open the qarm_mini_quick_start.slx Simulink model.



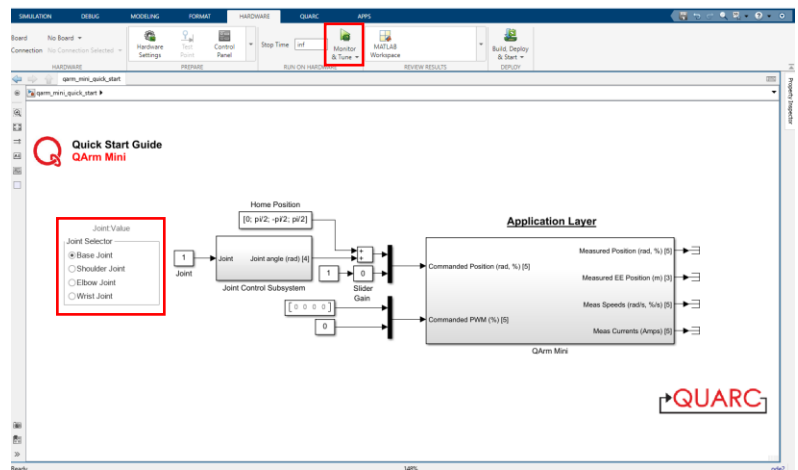
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1. From the root level's Application Layer, double-click QArm Mini subsystem to navigate to the Interface Layer, and then the Hardware Layer, and double-click on the HIL Initialize block.
2. Update the Board Identifier value to match the COM port you noted during setup (step 3A).



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1. Go back to the root level of your model (Application Layer). Build and deploy the model using the Monitor and Tune button on the Hardware or QUARC tab.
2. The manipulator starts in the home position. You can control each joint individually by selecting from the Joint Selector and using the up and down arrow keys on your keyboard.
3. You can press spacebar at anytime to return the arm to the home position.



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Stop the model. While supporting the QArm Mini manipulator by hand, turn OFF the power switch. The manipulator should now be gently moved to the Rest Configuration.

TROUBLESHOOTING

Common issues and possible solutions

QArm Mini Power switch
LED does not light up

Ensure that the power connector is connected firmly. Ensure that the USB-A to USB-C cable is connected to the manipulator and your workstation properly.



LEARN MORE

To browse and download the latest Quanser resources visit www.quanser.com/courseware or call +1-905-940-3575

STILL NEED HELP

For further assistance from a Quanser engineer, contact us at tech@quanser.com

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